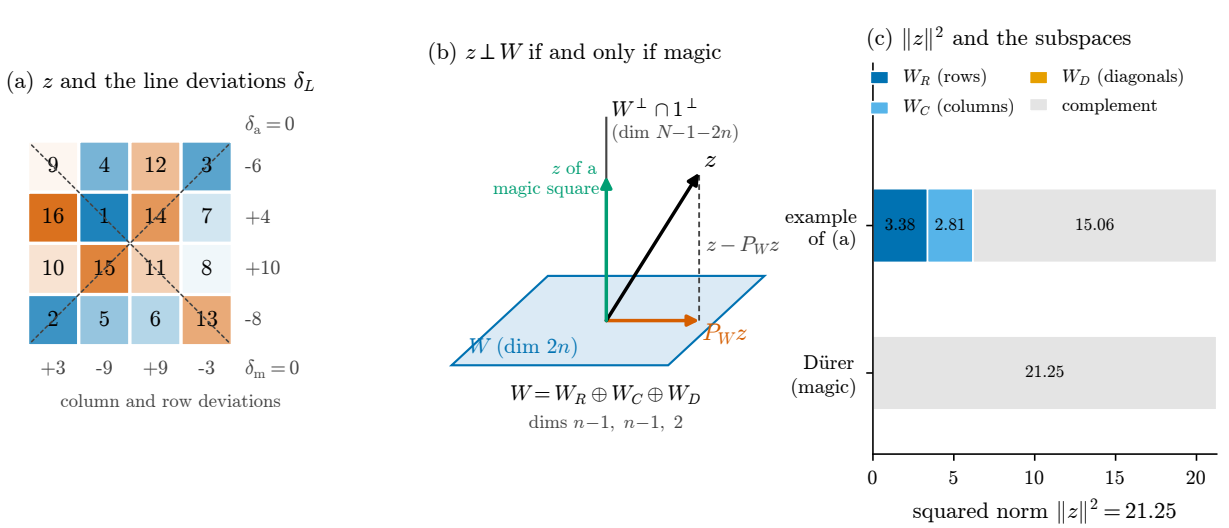
(a) z and the line deviations δ_L
$\delta_a = 0$
9 4 12 3 -6
16 1 14 7 +4
10 15 11 8 +10
2 5 6 13 -8
+3 -9 +9 -3 $\delta_m = 0$
column and row deviations
(b) $z \perp W$ if and only if magic
$W^\perp \cap 1^\perp$
(dim $N-1-2n$)
z of a magic square
z
$z - P_W z$
$P_W z$
W (dim $2n$)
$W = W_R \oplus W_C \oplus W_D$
dims $n-1$, $n-1$, 2
(c) $\|z\|^2$ and the subspaces
W_R (rows)
W_D (diagonals)
W_C (columns)
complement
example of (a)
3.38
2.81
15.06
Dürer (magic)
21.25
0 5 10 15 20
squared norm $\|z\|^2 = 21.25$